SEMITEACH IGBT



IGBT Module stack

SEMITEACH - IGBT
3-phase retifier + IGBT inverter + brake chopper

datasheet

Ordering No. 08753450
Description SEMITEACH IGBT
SKM50GB12T4, SKHI22A, SKD51/14

Features

- Multi-function IGBT converter
- IP2x protection for safety hazards
- Transparent enclosure to allow visualisation of internal part
- External connector for easy wiring
- Built in isolated IGBT driver and IGBT protection
- Forced-air cooled heatsink

Typical Applications

- Education : various converter configuration possible :

- 3-phase inverter+brake chopper
- Buck or boost converter
- single phase inverter
- single or 3-phase rectifier

Footnotes

1) The user shall ensure air ventilation for proper cooling

Remarks

This technical information specifies semiconductor devices but promises no characteristics. No warranty or guarantee, expressed or implied is made regarding delivery, performance or suitability.

Absolute maximum ratings			
Symbol	**Conditions**	**Values**	**Unit**
$I_{OUT\ MAX}$	Maximum permanent output current	**30**	A_{RMS}
$I_{IN\ MAX}$	Maximum permanent input current	**30**	A_{DC}
$V_{OUT\ MAX}$	Maximum output voltage	**400**	V_{AC}
$V_{BUS\ MAX}$	Maximum DC bus voltage	**750**	V_{DC}
f_{OUT}	Maximum inverter output frequency	**500**	Hz
f_{SW}	Maximum switching frequency	**50**	kHz

Electrical characteristics / Typical application			$T_{AIR\ COOLING\ 1)}$ = 30℃ unless otherwise specified			
Symbol	**Conditions**		**min**	**typ**	**max**	**Unit**
Ratings						
$I_{OUT\ RATED}$	Rated output current	No overload, tamb = 30℃ Chip junction T°< 150℃, (Max junction temperature = 175℃) With SEMIKRON axial fan assembly		**30**		A_{RMS}
V_{OUT}	Rated output voltage			**400**		V_{AC}
PF	Power factor			**1**		-
P_{OUT}	Rated output power			**20**		kW
f_{SW}	Inverter switching frequency			**5**		kHz
f_{OUT}	Output frequency			**50**		Hz
V_{BUS}	Rated DC voltage			**750**		V_{DC}
$P_{LOSS\ INV}$	Total power losses			**700**		W
η	Inverter efficiency			**-**		%

Protection & measurement					
Symbol	**Conditions**	**min**	**typ**	**max**	**Unit**
Thermal trip	Temperature trip level (Normally Open type: NO)		**71**		℃
Temperature sensing $T_{analogue\ OUT}$	Scaling over 30℃…110℃ temperature range				$mV.°C^{-1}$
	Linear temperature range	**30**		**110**	℃
	Accuracy of analogue signal over 65℃…110℃ range	**-1,5**		**1,5**	℃
	Max. output current			**5**	mA
	Max. voltage range	**0**		**10**	V_{DC}

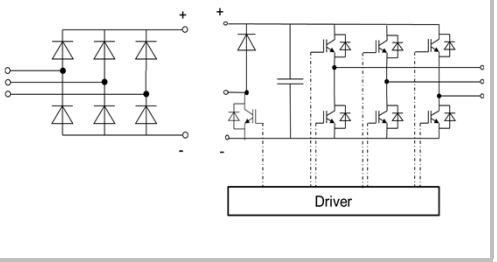
Axial fan data					
Heatsink fans	V_{SUPPLY}	Heatsink fan DC voltage supply		**230**	Vac
	P_{FAN}	Rated power at V_{SUPPLY} per fan, PWM 100%		**15**	W

Filtering characteristics					
V_{BUS}	Rated DC voltage applied to the caps bank with switching		**540**	**700**	V_{DC}
$V_{DC\ CAP}$	Max DC voltage applied to the caps bank without switching			**800**	V_{DC}
$\tau_{d5\%}$	Discharge time of the capacitors (5%)		**-**		s
C_{DC}	Capacitor bank capacity	**0,88**		**1,32**	mF
LTE	Calculated LTE of the caps with forced air cooling		**-**		kH

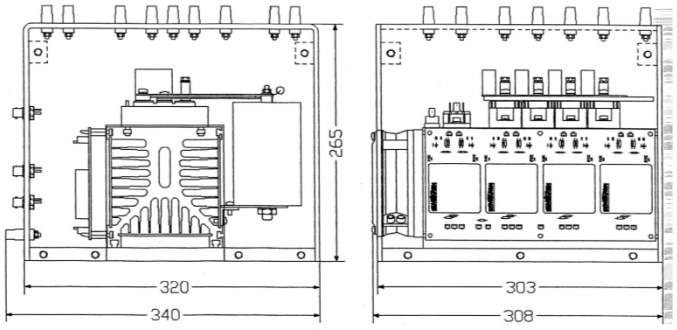
Stack Insulation				
V_{ISOL}	Frame / Power stage AC/DC (insulation test voltage AC, 60s)		**1 500**	V

Driver Characteristcs					
Symbol	**Conditions**	**min**	**typ**	**max**	**Unit**
Driver board data					
V_S	Supply voltage	**14,4**	**15**	**15,6**	V_{DC}
$I_{VP,\ IDLE}$	Supply primary current (no load)		**20**		mA
$I_{VP,\ LOAD}$	Max. supply primary current			**290**	mA
ViT+	input threshold voltage HIGH			**12,5**	V_{DC}
ViT-	input threshold voltage LOW	**4,5**			V_{DC}
R_{IN}	Input resistance		**10**		kΩ

Weight	3-phase IGBT inverter	**13,3**	kg
	3-phase IGBT inverter including fan assembly	**14,9**	



B6U + B6CI + E1CIKF



General dimensions